Hamiltonian of Mean Force Beyond the Commuting Gaussian Benchmark

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We study the operator representability problem for the Hamiltonian of mean force (HMF) in Gaussian linearly coupled environments beyond the commuting limit. Building on the quenched-density construction, we derive an exact imaginary-time reformulation in which bath statistics enter as scalar kernel moments while operator growth is generated by the adjoint chain $\{\mathrm{ad}^n_{H_Q}(f)\}_{n\geq 0}$. This yields a structural criterion: a finite closed-form HMF exists only when the adjoint-generated operator algebra closes inside the target ansatz. We then instantiate the criterion in a Pauli-closed qubit model and obtain an explicit closed-form mean-force Hamiltonian with temperature- and bath-dependence carried by mode-resolved kernel coefficients. The manuscript is framed as the noncommuting Gaussian follow-up to the commuting benchmark analysis of arXiv:2602.13146 and sets up the non-Gaussian extension program.

INTRODUCTION

This manuscript is a direct follow-up to Ref. [1] (arXiv:2602.13146). That paper established the quenched-density representation and delivered an exact benchmark in the commuting Gaussian sector. The present paper starts from that benchmark and addresses the next step: how the Hamiltonian of mean force is organised once the coupling operator does *not* commute with the bare system Hamiltonian.

In closed quantum statistical mechanics, equilibrium is generated by a Hamiltonian: $\rho \propto e^{-\beta H}$. For an open system with finite coupling, the operationally defined equilibrium state of the subsystem is the reduced state of the global Gibbs ensemble,

$$\bar{\rho}_S(\beta) = \mathrm{Tr}_B\, e^{-\beta H_{\mathrm{tot}}}, \tag{1}$$

$$e^{-\beta H_{\mathrm{MF}}(\beta)} \propto \frac{\mathrm{Tr}_B e^{-\beta H_{\mathrm{tot}}}}{Z_B(\beta)}, \quad Z_B(\beta) = \mathrm{Tr}_B e^{-\beta H_X}. \tag{2}$$

This object is generally not $e^{-\beta H_Q}$ for the bare system Hamiltonian H_Q. The resulting representational question is precise: what operator, if any, plays the role of an equilibrium generator for the subsystem once the coupling is non-negligible? The Hamiltonian of mean force (HMF) answers this by construction and is the standard starting point in strong-coupling thermodynamics and the mean-force Gibbs-state program [2–5].

Why care about a mean-force Hamiltonian at all? First, it provides the exact reduced equilibrium object that underlies strong-coupling thermodynamic identities, including free-energy and work relations that remain valid beyond weak coupling[2, 5, 6]. Second, it furnishes a consistent equilibrium initialization for open-system dynamics when correlations with the bath are unavoidable, a point emphasized in the literature on correlated initial states and reduced dynamics [4, 7]. Related subensemble approaches explicitly treat the subsystem thermodynamics as inherited from a global canonical ensemble, making the effective reduced generator central to the formalism[8]. In short, the HMF is not an optional reinterpretation; it is the exact operator that encodes the reduced equilibrium state whenever system–bath coupling is finite.

Historically, open-system theory prioritized weak-coupling and Markovian regimes, where reduced equilibrium can often be approximated by a Gibbs state of a renormalized H_Q. At finite coupling the reduced state inherits explicit temperature dependence and interaction-induced operator content that is not captured by a simple renormalization. Coupling-dependent thermodynamic response features in quantum Brownian motion and related models highlight this complexity[9, 10]. The strong-coupling literature consequently treats the mean-force Gibbs state as a distinct equilibrium object, with operational ramifications for heat and energy definitions[11, 12].

The HMF literature itself is broad but structured. Canonical definitions and thermodynamic identities are developed in strong-coupling thermodynamics and fluctuation-relation work[2, 3, 5, 6]. A comprehensive review consolidates the "static" mean-force Gibbs perspective with the "dynamical" return-to-equilibrium perspective in open quantum systems [4]. Operational questions such as measurability and thermodynamic consistency at strong coupling have also been pursued[12, 13].

Exact or controlled evaluations exist in special cases. For commuting (QND) interactions, the operator algebra closes trivially and the HMF can be written explicitly[14]. Quadratic/Gaussian models (e.g., damped harmonic oscillators) are solvable because Gaussianity is preserved, leading to closed operator forms[15–17]. Finite-dimensional closures such as spin-boson or single-qubit models provide additional controlled benchmarks[18]. Beyond these cases, the mean-force Gibbs state is often accessed through systematic limits: Cresser and Anders derive weak- and ultrastrong-coupling expressions and show that, in the ultrastrong limit, the mean-force Gibbs state becomes diagonal in the interaction basis rather than the system Hamiltonian basis[19]. Recent

work generalizes the HMF framework to finite baths by introducing a pair of quantum Hamiltonians of mean force that incorporate bath feedback[20], and structural studies further analyze the operator content of the HMF in extended settings[21].

Outside solvable models, most approaches are perturbative or numerical. Weak-coupling/high-temperature expansions yield controlled but limited series[19]. Imaginary-time path-integral methods, stochastic representations, and hierarchical-equations techniques can compute $\rho_S(\beta)$ directly but typically do not provide a compact operator form for H_{MF}[22–26]. Stochastic Liouville and partition-free approaches provide complementary numerical access to open-system equilibration without yielding closed-form HMFs [27, 28]. These methods are indispensable for quantitative predictions but leave open the representational question: when does a local or otherwise restricted operator form exist?

The influence-functional formalism is a natural language for this problem. For Gaussian baths with linear coupling it provides an exact route to integrating out bath degrees of freedom[15, 16, 29]. In equilibrium it becomes a Euclidean (imaginary-time) influence functional, usually bilocal in τ, and admits a Hubbard–Stratonovich rewriting as a quenched Gaussian-field average[30–32]. Related path-integral derivations of quantum Langevin dynamics make explicit the noise and dissipation structure inherited from the bath[33, 34]. This formulation also clarifies the terminology: "nonlocal" initially refers to imaginary-time nonlocality of the kernel, whereas the HMF locality question concerns the operator structure on the system Hilbert space.

The existence of H_{MF} is therefore not the issue—it is defined by a logarithm of a traced exponential. The real obstruction is representability: when does H_{MF} admit a closed-form expression within a restricted operator family (few-body, spatially local, or a given algebra)? Much of the literature either (i) solves special models, (ii) expands in controlled limits, or (iii) computes $\rho_S(\beta)$ numerically and analyzes its properties, but a general structural criterion is still lacking [19–21].

Relative to Ref. [1], the present manuscript has a narrower but deeper scope. We retain the exact Gaussian influence-functional and quenched-field identities, then use adjoint-action and Magnus/BCH organisation to study representability in the noncommuting sector. The central result is an explicit closure criterion: a finite closed-form HMF exists only when the operator algebra generated by repeated adjoint action of H_Q on the coupling operators closes inside the target ansatz. We then instantiate this criterion in a Pauli-closed qubit example, where all bath dependence is reduced to scalar kernel moments and the mean-force Hamiltonian is written in exact closed form.

MODEL

How do we model an environment? The Hamiltonian of mean force is defined by tracing the joint equilibrium operator, so its very definition requires a model for the composite Hamiltonian. To do so, we describe the subsystem by a Hamiltonian H_Q with coordinates q, coupled to a bath described by H_X (using x as coordinates) via an interaction H_I. The total system is then

$$H_{\mathrm{tot}} = H_Q + H_X + H_I. \tag{3}$$

Such a form is of course too general to be useful, and it is therefore necessary to specify a functional form for both the bath and the interaction. We choose a Caldeira-Leggett (CL) environment. This is appropriate when the bath is large and close to equilibrium, so its fluctuations are approximately Gaussian and the leading, linear coupling dominates a local expansion of the interaction. The bath is modeled as a collection of harmonic oscillators,

$$H_X = \sum_k \left(\frac{p_k^2}{2m_k} + \frac{1}{2} m_k \omega_k^2 x_k^2 \right). \tag{4}$$

We then specify the interaction as

$$H_I = \sum_k c_k f(q) x_k, \tag{5}$$

where the bath coordinate x_k is coupled with strength c_k to a system operator $f \equiv f(q)$. The total Hamiltonian is then

$$H_{\mathrm{tot}} = H_Q + \sum_k \left(\frac{p_k^2}{2m_k} + \frac{1}{2} m_k \omega_k^2 x_k^2 + c_k f(q) x_k \right). \tag{6}$$

To enforce translational invariance, CL models often insert a counter-term proportional to $f(q)^2$. This cancels the potential renormalisation induced by the coupling, and may be absorbed into the definition of H_Q. This framework generalizes the Jaynes-Cummings model to a multi-mode environment, reducing to it when the system is a two-level atom and a single mode is considered.

Finally, while the bath is parametrised by the masses m_k, frequencies ω_k, and coupling strengths c_k, its influence on the system may ultimately be characterised by the spectral density $J(\omega)$, defined by

$$J(\omega) = \frac{\pi}{2} \sum_k \frac{c_k^2}{m_k \omega_k} \delta(\omega - \omega_k). \tag{7}$$

This compactly characterises the bath spectral properties and ultimately determines the form of bath correlations with the system.

Hamiltonian of mean force and the obstruction to a direct construction

With the composite Hamiltonian specified, we turn our attention to the central question of this paper - the con-

struction of the *Hamiltonian of mean force* $H_{\rm MF}$. This serves as the effective thermodynamic description of the reduced system, and it is defined as the operator whose Gibbs form reproduces the reduced equilibrium object up to a chosen normalisation. If the total system is in equilibrium at inverse temperature β, it will be described (up to normalisation) by $e^{-\beta H_{\rm tot}}$. The subsystem, however, does not inherit a Gibbs form generated by H_Q alone. The correct reduced equilibrium object is instead

$$\bar{\rho}_S(\beta) \equiv \mathrm{Tr}_X\, e^{-\beta H_{\rm tot}}, \tag{8}$$

where Tr_X traces out the bath degrees of freedom. The Hamiltonian of mean force is defined as the operator whose Gibbs form reproduces this reduced object up to a chosen normalisation:

$$e^{-\beta H_{\rm MF}(\beta)} \equiv \frac{\mathrm{Tr}_X\, e^{-\beta H_{\rm tot}}}{Z_X(\beta)}, \qquad Z_X(\beta) \equiv \mathrm{Tr}_X\, e^{-\beta H_X}. \tag{9}$$

Equivalently, $H_{\rm MF}(\beta) = -(1/\beta)\log\big[\mathrm{Tr}_X e^{-\beta H_{\rm tot}}\big]$ up to an additive scalar fixed by the choice of Z_X.

At first sight one might hope to evaluate the trace in Eq. (9) directly. For a harmonic bath linearly coupled through a collective coordinate, the bath can indeed be eliminated exactly, yielding the Feynman–Vernon influence functional [29]. This formalism has been used to obtain exact reduced descriptions at the level of stochastic equations of motion in a variety of settings [16, 32, 35]. However, this dynamical tractability does not translate into a tractable static expression for $H_{\rm MF}$. The obstruction is structural: eliminating the bath produces a bilocal self-interaction in imaginary time, with a kernel fixed by the bath two-point structure (equivalently by $J(\omega)$), acting on the interaction-picture operator associated with the coupling f. When $[H_Q, f] \neq 0$, this yields a time-ordered exponential of a memory-kernel self-coupling [2, 8], rather than a simple exponential $e^{-\beta H_{\rm eff}}$ of a local operator. Identifying the Hamiltonian of mean force—the logarithm of this ordered object—is therefore a non-trivial inverse problem.

The influence functional nevertheless contains exactly the ingredients required to construct $H_{\rm MF}(\beta)$ and to state conditions under which it closes within a restricted operator class. Our strategy is to rewrite the influence functional so that these ingredients are separated cleanly: bath statistics enter only through $J(\omega)$ (or the associated kernel), while all nontrivial operator content is isolated to the algebra generated by H_Q and f. To achieve this, it is necessary to first connect the imaginary-time propagator of the reduced system to a path integral representation of the influence functional.

QUENCHED REPRESENTATION OF THE REDUCED EQUILIBRIUM OPERATOR

The Hamiltonian of mean force is defined implicitly by the reduced equilibrium operator $\bar{\rho}_S(\beta) = \mathrm{Tr}_X e^{-\beta H_{\rm tot}}$. To obtain a constructive handle on $H_{\rm MF}(\beta)$, it is useful to reorganise the problem in two steps.

First, we treat $\bar{\rho}_S(\beta)$ as an imaginary-time propagator on the system Hilbert space: it is a positive operator obtained by evolving over an interval of length β, and any such operator may be written in the form

$$\bar{\rho}_S(\beta) = \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\, H_{\rm eff}(\tau)\right], \tag{10}$$

for some (in general non-unique) generator $H_{\rm eff}(\tau)$.

Second, we explicitly allow the generator to depend on τ. This is not a technical luxury but a structural necessity: after eliminating the bath, the system acquires a non-local memory-kernel self-coupling. As emphasized in recent partition-free approaches [36, 37] and the development of stochastic Liouville-von Neumann methods [27, 32, 38, 39], this non-local object can be mapped *exactly* to a local, stochastic evolution via the Hubbard-Stratonovich transformation. This transformation trades the temporal non-locality for an average over an auxiliary Gaussian field, ensuring that the full non-Markovian character of the bath is retained without approximation.

A full derivation of this result is provided in Appendix .

For a CL environment the bath can be eliminated exactly, yielding an imaginary-time influence functional with a memory kernel fixed by the spectral density $J(\omega)$. Rather than working with the resulting bilocal object directly, we adopt an equivalent representation in which the bath enters through an auxiliary Gaussian field and the system evolves with a *local-in-τ* generator. Specifically, we introduce a real stochastic field $\xi(\tau)$ on $\tau \in [0, \beta]$ with

$$\langle \xi(\tau)\rangle_\xi = 0, \qquad \langle \xi(\tau)\xi(\tau')\rangle_\xi = K(\tau - \tau'), \tag{11}$$

where K is the Euclidean bath correlation kernel determined by $J(\omega)$. A convenient explicit form is

$$K(\tau) = \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\, \frac{\cosh\big(\omega(\beta/2 - |\tau|)\big)}{\sinh(\beta\omega/2)}, \qquad \tau \in [-\beta, \beta]. \tag{12}$$

For each realisation ξ, we define the quenched effective Hamiltonian

$$H_{\rm eff}[\xi](\tau) \equiv H_Q + \xi(\tau)\, f, \tag{13}$$

and the corresponding quenched propagator

$$U_\xi(\beta) \equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\, H_{\rm eff}[\xi](\tau)\right]. \tag{14}$$

The reduced equilibrium operator admits the quenched average representation

$$\bar{\rho}_S(\beta) \equiv \mathrm{Tr}_X e^{-\beta H_{\rm tot}} \;=\; \big\langle U_\xi(\beta)\big\rangle_\xi, \tag{15}$$

with the Gaussian measure on ξ fixed by Eqs.(11)–(12). Consequently, performing the average in (15) yields the Hamiltonian of mean force identically via

$$H_{\rm MF}(\beta) = -\frac{1}{\beta}\log\left[\frac{\bar\rho_S(\beta)}{Z_X(\beta)}\right], \qquad Z_X(\beta) = {\rm Tr}_X e^{-\beta H_X}, \tag{16}$$

with the scalar fixed by the chosen normalisation. Consequently, calculating $H_{\rm MF}(\beta)$ is equivalent to determining $\bar\rho_S(\beta)$, which in turn requires performing the average in (15) over the Gaussian field ξ.

To perform this trick, it is necessary first to introduce the imaginary-time interaction picture with respect to H_Q by writing

$$\begin{aligned} U_\xi(\beta) &= e^{-\beta H_Q}\, W_\xi(\beta), \\ W_\xi(\beta) &\equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\, \xi(\tau)\, \tilde f(\tau)\right], \end{aligned} \tag{17}$$

where

$$\tilde f(\tau) \equiv e^{\tau H_Q}\, f\, e^{-\tau H_Q}. \tag{18}$$

This factors out the $\int H_Q$ piece from the ordered exponential, leaving $\int \xi(\tau)\tilde f(\tau)$ as the remaining term to be averaged over. As a final step before averaging, it is convenient to introduce the adjoint action of H_Q on f, defined as ${\rm ad}_{H_Q}(f) = [H_Q, f]$ and ${\rm ad}^n_{H_Q}(f) = [H_Q, {\rm ad}^{n-1}_{H_Q}(f)]$. Then the interaction-picture coupling obeys

$$\tilde f(\tau) = e^{\tau H_Q} f e^{-\tau H_Q} = e^{\tau\, {\rm ad}_{H_Q}}(f) = \sum_{n=0}^\infty \frac{\tau^n}{n!}\, {\rm ad}^n_{H_Q}(f). \tag{19}$$

All operator growth induced by eliminating the bath is therefore controlled by the nested-commutator chain $\{{\rm ad}^n_{H_Q}(f)\}_{n\geq 0}$.

We now compute the averaged interaction-picture propagator and actor out the free evolution:

$$U_\xi(\beta) = e^{-\beta H_Q}\, W_\xi(\beta), \qquad W_\xi(\beta) \equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\, \xi(\tau)\, \tilde f(\tau)\right]. \tag{20}$$

Taking the Gaussian average yields

$$\bar\rho_S(\beta) = e^{-\beta H_Q}\, \bar W(\beta), \qquad \bar W(\beta) \equiv \big\langle W_\xi(\beta)\big\rangle_\xi. \tag{21}$$

Thus the task is to evaluate the averaged interaction-picture propagator $\bar W(\beta)$. To do so, we expand the time-ordered exponential in Eq. (20) in the standard Dyson series form,

$$\begin{aligned} W_\xi(\beta) = \sum_{k=0}^\infty (-1)^k \int_{\Delta_k} d\tau_1\cdots d\tau_k\, \xi(\tau_1)\cdots\xi(\tau_k) \\ \times \tilde f(\tau_1)\cdots\tilde f(\tau_k), \end{aligned} \tag{22}$$

where Δ_k denotes the time-ordered integration domain

$$\Delta_k: \qquad 0 \leq \tau_k \leq \cdots \leq \tau_2 \leq \tau_1 \leq \beta, \tag{23}$$

and $d\tau_1\cdots d\tau_k$ is shorthand for the product measure over that domain.

Averaging term-by-term gives

$$\begin{aligned} \bar W(\beta) = \sum_{k=0}^\infty (-1)^k \int_{\Delta_k} d\tau_1\cdots d\tau_k\, \big\langle \xi(\tau_1)\cdots\xi(\tau_k)\big\rangle_\xi \\ \times \tilde f(\tau_1)\cdots\tilde f(\tau_k). \end{aligned} \tag{24}$$

Gaussian moments and Wick's theorem. Since ξ is Gaussian with zero mean, all odd moments vanish:

$$\big\langle \xi(\tau_1)\cdots\xi(\tau_{2n+1})\big\rangle_\xi = 0. \tag{25}$$

For even moments, Wick's theorem (Isserlis' theorem) yields a sum over all pairings:

$$\big\langle \xi(\tau_1)\cdots\xi(\tau_{2n})\big\rangle_\xi = \sum_{P\in\mathcal{P}_{2n}} \prod_{(i,j)\in P} K(\tau_i - \tau_j), \tag{26}$$

where $\mathcal{P}_{2n}$ is the set of perfect matchings (pair partitions) of $\{1,\ldots,2n\}$. Substituting Eq. (26) into Eq. (24) gives the explicit all-orders expansion

$$\begin{aligned} \bar W(\beta) = \sum_{n=0}^\infty \int_{\Delta_{2n}} d\tau_1\cdots d\tau_{2n} \left(\sum_{P\in\mathcal{P}_{2n}} \prod_{(i,j)\in P} K(\tau_i - \tau_j)\right) \\ \times \tilde f(\tau_1)\cdots\tilde f(\tau_{2n}). \end{aligned} \tag{27}$$

This series is precisely the Wick expansion of a time-ordered exponential with a quadratic (pairwise) contraction. Since the exponent in Eq. (20) is linear in the Gaussian field, the average can be resummed exactly to yield the bilocal influence-functional form

$$\boxed{\bar W(\beta) = \mathcal{T}_\tau \exp\left[\frac{1}{2}\int_0^\beta d\tau \int_0^\beta d\tau'\; K(\tau-\tau')\, \tilde f(\tau)\tilde f(\tau')\right].} \tag{28}$$

For later convenience we define the bilocal quadratic operator

$$\Delta \;\equiv\; \frac{1}{2}\int_0^\beta d\tau \int_0^\beta d\tau'\; K(\tau-\tau')\, \tilde f(\tau)\tilde f(\tau'), \tag{29}$$

so that

$$\bar W(\beta) = \mathcal{T}_\tau e^{\Delta}. \tag{30}$$

At this stage, the Gaussian field has been eliminated exactly, and all bath dependence is contained in the translation-invariant kernel $K(\tau-\tau')$. In the next step we exploit this translation invariance to diagonalise K in Matsubara modes and rewrite Δ as a sum of squares of mode-projected operators.

Diagonalising the translation-invariant kernel and a sum-of-squares form

We now exploit the key structural property of the Euclidean correlation kernel: it is translation invariant on the thermal circle, i.e. it depends only on the time difference. Equivalently, K is β-periodic and admits a bosonic Matsubara Fourier series. This allows us to rewrite the bilocal quadratic operator Δ in Eq. (29) in a form that isolates all bath dependence into scalar mode weights.

Matsubara expansion of the kernel. On $\tau \in [0, \beta]$, any β-periodic translation-invariant kernel may be expanded as

$$K(u) = \frac{1}{\beta} \sum_{\ell \in \mathbb{Z}} \kappa_\ell \, e^{i\nu_\ell u}, \qquad \nu_\ell \equiv \frac{2\pi\ell}{\beta}, \tag{31}$$

with Fourier coefficients

$$\kappa_\ell \equiv \int_0^\beta du \; K(u) \, e^{-i\nu_\ell u}. \tag{32}$$

For a real, even kernel $K(u) = K(-u)$ one has $\kappa_\ell \in \mathbb{R}$ and $\kappa_{-\ell} = \kappa_\ell$. (For the Caldeira–Leggett class one further has $\kappa_\ell \geq 0$.)

Substituting Eq. (31) into Eq. (29) gives

$$\begin{aligned} \Delta &= \frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' \left[\frac{1}{\beta} \sum_{\ell \in \mathbb{Z}} \kappa_\ell \, e^{i\nu_\ell(\tau - \tau')} \right] \tilde{f}(\tau) \tilde{f}(\tau') \\ &= \frac{1}{2\beta} \sum_{\ell \in \mathbb{Z}} \kappa_\ell \left(\int_0^\beta d\tau \, e^{i\nu_\ell \tau} \, \tilde{f}(\tau) \right) \left(\int_0^\beta d\tau' \, e^{-i\nu_\ell \tau'} \, \tilde{f}(\tau') \right). \end{aligned} \tag{33}$$

Mode-projected operators and sum-of-squares form. Define the Matsubara-projected interaction-picture operators

$$\tilde{F}_\ell \;\equiv\; \int_0^\beta d\tau \, e^{i\nu_\ell \tau} \, \tilde{f}(\tau). \tag{34}$$

Since $\tilde{f}(\tau)$ is Hermitian for Hermitian f, one has $\tilde{F}_{-\ell} = \tilde{F}_\ell^\dagger$. Equation (33) then becomes the exact quadratic sum

$$\boxed{\Delta = \frac{1}{2\beta} \sum_{\ell \in \mathbb{Z}} \kappa_\ell \, \tilde{F}_\ell \, \tilde{F}_\ell^\dagger.} \tag{35}$$

This form cleanly separates bath and system structure: all information about the bath enters through the scalar mode weights $\{\kappa_\ell\}$, while the system algebra is carried by the family of operators $\{\tilde{F}_\ell\}$.

Manifestly real form (optional). When K is real and even (so $\kappa_\ell = \kappa_{-\ell} \in \mathbb{R}$), it is often convenient to pair $\pm\ell$ and write Δ in terms of cosine and sine projections. Using

$$K(u) = \frac{\kappa_0}{\beta} + \frac{2}{\beta} \sum_{\ell \geq 1} \kappa_\ell \cos(\nu_\ell u), \tag{36}$$

define the real mode operators

$$\begin{aligned} \tilde{F}_{\ell,c} &\equiv \int_0^\beta d\tau \; \cos(\nu_\ell \tau) \, \tilde{f}(\tau), \\ \tilde{F}_{\ell,s} &\equiv \int_0^\beta d\tau \; \sin(\nu_\ell \tau) \, \tilde{f}(\tau), \quad (\ell \geq 1), \end{aligned} \tag{37}$$

and $\tilde{F}_{0,c} \equiv \int_0^\beta d\tau \, \tilde{f}(\tau)$. Then Eq. (35) is equivalent to

$$\Delta = \frac{\kappa_0}{2\beta} \, \tilde{F}_{0,c}^2 + \frac{1}{2\beta} \sum_{\ell \geq 1} \kappa_\ell \left(\tilde{F}_{\ell,c}^2 + \tilde{F}_{\ell,s}^2 \right), \tag{38}$$

making explicit that the quadratic form is real.

In the next step we express the projected operators $\tilde{F}_\ell$ (or equivalently $\tilde{F}_{\ell,c/s}$) in terms of the nested-commutator chain generated by H_Q acting on f. This yields a generating-function representation for all kernel moments and will allow us to collect the bath dependence into scalar prefactors multiplying products of $\mathrm{ad}_{H_Q}^n(f)$.

Adjoint-chain expansion and generating functions for Matsubara moments

We now rewrite the mode-projected operators $\tilde{F}_\ell$ in Eq. (34) in a basis that makes the operator growth induced by the bath completely explicit. The key observation is that the entire τ-dependence of $\tilde{f}(\tau)$ is generated by the adjoint action of H_Q.

Adjoint chain and interaction-picture coupling. Introduce the adjoint superoperator

$$\begin{aligned} &\mathrm{ad}_{H_Q}(A) \equiv [H_Q, A], \quad \mathrm{ad}_{H_Q}^0(A) \equiv A, \\ &\mathrm{ad}_{H_Q}^n(A) \equiv [H_Q, \mathrm{ad}_{H_Q}^{n-1}(A)]. \end{aligned} \tag{39}$$

Define the nested-commutator chain generated from f,

$$f_n \;\equiv\; \mathrm{ad}_{H_Q}^n(f), \qquad n \geq 0. \tag{40}$$

Then the interaction-picture operator is

$$\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q} = e^{\tau \, \mathrm{ad}_{H_Q}}(f) = \sum_{n=0}^\infty \frac{\tau^n}{n!} \, f_n. \tag{41}$$

Equation (41) cleanly separates (i) the scalar monomials $\tau^n/n!$ from (ii) the operator content f_n, and the latter is the sole source of operator growth.

Matsubara moments and operator-valued mode expansion. Substituting Eq. (41) into the mode definition (34) yields

$$\tilde{F}_\ell = \int_0^\beta d\tau \, e^{i\nu_\ell \tau} \sum_{n=0}^\infty \frac{\tau^n}{n!} \, f_n = \sum_{n=0}^\infty I_n(\nu_\ell) \, f_n, \tag{42}$$

where we have defined the factorial-weighted Matsubara moments

$$I_n(\nu) \;\equiv\; \int_0^\beta d\tau \, \frac{\tau^n}{n!} \, e^{i\nu\tau}. \tag{43}$$

Since $\nu_{-\ell} = -\nu_\ell$, one has $I_n(\nu_{-\ell}) = I_n(\nu_\ell)^*$ and hence $\tilde{F}_{-\ell} = \tilde{F}_\ell^\dagger$ for Hermitian f.

Generating function for all moments $I_n(\nu_\ell)$. The full family $\{I_n(\nu_\ell)\}_{n\geq 0}$ is generated by the scalar function

$$I(\nu;s) \;\equiv\; \sum_{n=0}^{\infty} s^n\, I_n(\nu) = \int_0^\beta d\tau\, \exp\big[(s+i\nu)\tau\big] = \frac{e^{(s+i\nu)\beta}-1}{s+i\nu}. \tag{44}$$

For bosonic Matsubara frequencies $\nu = \nu_\ell = 2\pi\ell/\beta$ one has $e^{i\nu_\ell\beta} = 1$, and therefore

$$I(\nu_\ell;s) = \frac{e^{s\beta}-1}{s+i\nu_\ell}. \tag{45}$$

The individual coefficients are recovered by differentiation at $s=0$:

$$I_n(\nu_\ell) = \frac{1}{n!}\,\partial_s^n I(\nu_\ell;s)\Big|_{s=0}. \tag{46}$$

Equation (45) provides a compact representation for all prefactors produced by the time integrals.

Factorisation of the quadratic form. Combining Eq. (35) with Eq. (42) yields

$$\Delta = \frac{1}{2\beta}\sum_{\ell\in\mathbb{Z}} \kappa_\ell \left(\sum_{n=0}^{\infty} I_n(\nu_\ell)\, f_n\right)\left(\sum_{m=0}^{\infty} I_m(\nu_\ell)^*\, f_m\right) = \frac{1}{2}\sum_{n,m=0}^{\infty} C_{nm}(\beta)\, f_n f_m, \tag{47}$$

where the moment matrix is

$$C_{nm}(\beta) \;\equiv\; \frac{1}{\beta}\sum_{\ell\in\mathbb{Z}} \kappa_\ell\, I_n(\nu_\ell)\, I_m(\nu_\ell)^*. \tag{48}$$

This makes the desired decoupling explicit: C_{nm} factorises as a sum of outer products over Matsubara modes. Equivalently, defining mode coefficients

$$a_{n\ell} \;\equiv\; \sqrt{\frac{\kappa_\ell}{\beta}}\, I_n(\nu_\ell), \tag{49}$$

one has

$$C_{nm}(\beta) = \sum_{\ell\in\mathbb{Z}} a_{n\ell}\, a^*_{m\ell}, \tag{50}$$

and therefore

$$\Delta = \frac{1}{2}\sum_{\ell\in\mathbb{Z}} \left(\sum_{n\geq 0} a_{n\ell}\, f_n\right)\left(\sum_{m\geq 0} a^*_{m\ell}\, f_m\right). \tag{51}$$

All temperature and spectral-density dependence is confined to the scalar mode weights κ_ℓ and the generating function (45), while all operator complexity resides in the adjoint chain $\{f_n\}$.

In the next step we return from the interaction picture to the Schrödinger picture, combining $e^{-\beta H_Q}$ with the averaged propagator $\bar{W}(\beta) = \mathcal{T}_\tau e^{\Delta}$ to obtain a single-exponent representation and hence a systematic expansion of the mean-force Hamiltonian in commutator order.

De-interaction picture and single-exponent form

At this point the Gaussian average has been performed exactly, and the averaged interaction-picture propagator is

$$\bar{W}(\beta) = \mathcal{T}_\tau e^{\Delta}, \qquad \Delta = \frac{1}{2\beta}\sum_{\ell\in\mathbb{Z}} \kappa_\ell\, \tilde{F}_\ell \tilde{F}_\ell^\dagger = \frac{1}{2}\sum_{n,m\geq 0} C_{nm}(\beta)\, f_n f_m, \tag{52}$$

with $\tilde{F}_\ell = \sum_{n\geq 0} I_n(\nu_\ell) f_n$ and C_{nm} given by Eq. (48).

Time ordering becomes redundant after mode reduction. Although Eq. (52) is written as a time-ordered exponential, the operator Δ itself contains no remaining τ-dependence: all τ-integrals have been carried out explicitly into scalar coefficients (κ_ℓ, I_n, and hence C_{nm}), while the operator content is carried by the time-independent adjoint-chain elements $\{f_n\}$. Therefore $\mathcal{T}_\tau$ acts trivially and we may write

$$\bar{W}(\beta) = e^{\Delta}. \tag{53}$$

Consequently, the reduced equilibrium operator takes the nonperturbative product form

$$\bar{\rho}_S(\beta) = e^{-\beta H_Q}\, e^{\Delta}. \tag{54}$$

Definition of the effective (mean-force) Hamiltonian. We now define an effective system Hamiltonian $H_{\rm eff}(\beta)$ by the single-exponent representation

$$\bar{\rho}_S(\beta) \equiv e^{-\beta H_{\rm eff}(\beta)}, \tag{55}$$

up to the overall scalar normalisation fixed by $Z_X(\beta)$ in Eq. (16). Combining Eqs. (54) and (55) gives

$$-\beta H_{\rm eff}(\beta) = \log\big(e^{-\beta H_Q} e^{\Delta}\big), \tag{56}$$

so $H_{\rm eff}$ is obtained by the Baker–Campbell–Hausdorff (BCH) series with

$$A \equiv -\beta H_Q, \qquad B \equiv \Delta, \qquad \log(e^A e^B) = A + B + \frac{1}{2}[A,B] + \frac{1}{12}[A,[A,B]] - \frac{1}{12}[B,[A,B]] + \cdots. \tag{57}$$

Systematic commutator-order expansion at fixed quadratic content. A key advantage of the adjoint-chain representation is that commutators with H_Q simply shift the chain index. Indeed, by definition $[H_Q, f_n] = f_{n+1}$, and the adjoint action satisfies the Leibniz rule. Iterating yields the explicit binomial decomposition

$$\mathrm{ad}^r_{H_Q}(f_n f_m) = \sum_{k=0}^{r} \binom{r}{k} f_{n+k}\, f_{m+r-k}, \qquad r \geq 0, \tag{58}$$

which isolates the combinatorics (the binomial factors) from the operator content.

Using Eq. (47), the r-fold adjoint action of H_Q on Δ is therefore

$$\begin{aligned}\mathrm{ad}^r_{H_Q}(\Delta) &= \frac{1}{2}\sum_{n,m\geq 0} C_{nm}(\beta)\,\mathrm{ad}^r_{H_Q}(f_n f_m)\\ &= \frac{1}{2}\sum_{n,m\geq 0} C_{nm}(\beta)\sum_{k=0}^{r}\binom{r}{k} f_{n+k}\, f_{m+r-k}.\end{aligned} \tag{59}$$

Linear-in-Δ resummation (Bernoulli/BCH kernel). If we retain only terms *linear* in Δ in the BCH series (i.e. we drop terms of order Δ^2 and higher, such as $[B,[A,B]]$ in Eq. (57)), then the remaining commutator tower $B+\frac{1}{2}[A,B]+\frac{1}{12}[A,[A,B]]+\cdots$ can be resummed to all commutator orders via a Bernoulli series. Specifically,

$$\begin{aligned}\log(e^A e^B) &= A + \varphi(\mathrm{ad}_A)\,B + \mathcal{O}(B^2),\\ \varphi(x) &\equiv \frac{x}{1-e^{-x}} = \sum_{r=0}^{\infty}\frac{B_r}{r!}\,x^r,\end{aligned} \tag{60}$$

where B_r are Bernoulli numbers with the convention $B_0 = 1$ and $B_1 = +1/2$. Since $A = -\beta H_Q$, one has $\mathrm{ad}_A = -\beta\,\mathrm{ad}_{H_Q}$, and hence

$$-\beta H_{\mathrm{eff}}(\beta) = -\beta H_Q + \varphi\big(-\beta\,\mathrm{ad}_{H_Q}\big)\Delta + \mathcal{O}(\Delta^2). \tag{61}$$

Equivalently,

$$\boxed{H_{\mathrm{eff}}(\beta) = H_Q - \frac{1}{\beta}\,\varphi\big(-\beta\,\mathrm{ad}_{H_Q}\big)\Delta + \mathcal{O}(\Delta^2).} \tag{62}$$

Expanding φ explicitly gives the commutator-order series

$$H_{\mathrm{eff}}(\beta) = H_Q - \frac{1}{\beta}\sum_{r=0}^{\infty}\frac{B_r}{r!}\,(-\beta)^r\,\mathrm{ad}^r_{H_Q}(\Delta) + \mathcal{O}(\Delta^2), \tag{63}$$

and substituting Eq. (59) yields the fully explicit adjoint-chain form

$$\begin{aligned}H_{\mathrm{eff}}(\beta) = H_Q - \frac{1}{2\beta}\sum_{r=0}^{\infty}\frac{B_r}{r!}\,(-\beta)^r \sum_{n,m\geq 0} C_{nm}(\beta)\\ \times\sum_{k=0}^{r}\binom{r}{k} f_{n+k}\, f_{m+r-k} + \mathcal{O}(\Delta^2).\end{aligned} \tag{64}$$

All bath dependence is contained in the scalar matrix $C_{nm}(\beta)$ (or equivalently κ_ℓ and $I_n(\nu_\ell)$), while the full operator content is controlled by the adjoint chain $\{f_n\}$ and the binomial factors in Eq. (58).

Commuting baseline and commutator expansion of $H_{\mathrm{eff}}(\beta)$

It is useful to isolate the simplest limiting case in which the coupling operator f commutes with the system Hamiltonian H_Q. This provides both a consistency check and a natural "baseline" around which to organise the general noncommuting corrections in terms of nested commutators with H_Q.

Commuting check: $[H_Q, f] = 0$. If $[H_Q, f] = 0$, then the interaction-picture coupling is τ-independent: $\tilde{f}(\tau) = f$. The Matsubara-projected operators defined in Eq. (34) reduce to

$$\tilde{F}_\ell = \int_0^\beta d\tau\, e^{i\nu_\ell\tau}\, f = f\int_0^\beta d\tau\, e^{i\nu_\ell\tau} = \beta f\,\delta_{\ell 0}, \tag{65}$$

so only the $\ell = 0$ mode contributes to the sum-of-squares form (35). Consequently,

$$\Delta = \frac{1}{2\beta}\kappa_0\,(\beta f)(\beta f) = \frac{\beta\kappa_0}{2}\,f^2. \tag{66}$$

Since Δ then commutes with H_Q, the reduced equilibrium operator $\bar{\rho}_S(\beta) = e^{-\beta H_Q}e^{\Delta}$ collapses to a single exponential,

$$\bar{\rho}_S(\beta) = \exp\Big[-\beta\Big(H_Q - \frac{\kappa_0}{2}f^2\Big)\Big]. \tag{67}$$

Thus in the commuting limit the effective Hamiltonian acquires a purely local quadratic correction,

$$H_{\mathrm{eff}}(\beta) = H_Q - \frac{\kappa_0}{2}f^2, \qquad \text{when } [H_Q, f] = 0, \tag{68}$$

consistent with interpreting κ_0 as the zero-frequency weight of the Euclidean bath correlations.

General case: organise H_{eff} in f and commutators with H_Q. Away from the commuting limit, we retain the exact product form

$$\bar{\rho}_S(\beta) = e^{-\beta H_Q}e^{\Delta}, \tag{69}$$

with Δ given explicitly by Eqs. (35)–(47). We then define $H_{\mathrm{eff}}(\beta)$ by

$$\bar{\rho}_S(\beta) \equiv e^{-\beta H_{\mathrm{eff}}(\beta)}, \tag{70}$$

so that $-\beta H_{\mathrm{eff}}(\beta) = \log(e^{-\beta H_Q}e^{\Delta})$.

To express H_{eff} in a form that depends only on f and nested commutators with H_Q, we use the adjoint-chain representation $f_n \equiv \mathrm{ad}^n_{H_Q}(f)$ in Eq. (40). In particular, $[H_Q, f_n] = f_{n+1}$ and for products the r-fold adjoint action obeys the binomial identity

$$\mathrm{ad}^r_{H_Q}(f_n f_m) = \sum_{k=0}^{r}\binom{r}{k} f_{n+k}\, f_{m+r-k}. \tag{71}$$

Using Eq. (47), the commutator tower $\mathrm{ad}^r_{H_Q}(\Delta)$ is therefore

$$\mathrm{ad}^r_{H_Q}(\Delta) = \frac{1}{2}\sum_{n,m\geq 0} C_{nm}(\beta)\sum_{k=0}^{r}\binom{r}{k} f_{n+k}\, f_{m+r-k}. \tag{72}$$

This exhibits the desired separation: all β- and bath-dependence resides in the scalar coefficients $C_{nm}(\beta)$, while the operator basis is generated entirely by f and successive commutators with H_Q.

Linear-in-Δ commutator resummation. Retaining only terms linear in Δ in the BCH series for $\log(e^{-\beta H_Q} e^{\Delta})$, the commutator tower can be resummed to all orders in ad_{H_Q} using Bernoulli numbers:

$$H_{\mathrm{eff}}(\beta) = H_Q - \frac{1}{\beta} \sum_{r=0}^{\infty} \frac{B_r}{r!} (-\beta)^r \, \mathrm{ad}^r_{H_Q}(\Delta) + \mathcal{O}(\Delta^2), \quad (73)$$

with $B_0 = 1$ and $B_1 = +1/2$. Substituting Eq. (72) yields an explicit expansion in products of commutator-chain elements:

$$\boxed{\begin{aligned} H_{\mathrm{eff}}(\beta) = H_Q &- \frac{1}{2\beta} \sum_{r=0}^{\infty} \frac{B_r}{r!} (-\beta)^r \\ &\times \sum_{n,m\geq 0} C_{nm}(\beta) \sum_{k=0}^{r} \binom{r}{k} f_{n+k} \, f_{m+r-k} \\ &+ \mathcal{O}(\Delta^2). \end{aligned}} \quad (74)$$

Equation (74) is a systematic commutator-order construction of the effective Hamiltonian: for any truncation in the adjoint chain $\{f_n\}$ one obtains a closed operator approximation, with all scalar prefactors determined exactly from the translation-invariant kernel via Eqs. (48) and (45).

Normalisation and the Hamiltonian of mean force

The Hamiltonian of mean force is defined (up to an additive scalar multiple of the identity) by

$$H_{\mathrm{MF}}(\beta) = -\frac{1}{\beta} \log\left[\frac{\bar{\rho}_S(\beta)}{Z_X(\beta)}\right], \qquad Z_X(\beta) = \mathrm{Tr}_X e^{-\beta H_X}. \quad (75)$$

Using the definition of H_{eff} in Eq. (70),

$$\bar{\rho}_S(\beta) = e^{-\beta H_{\mathrm{eff}}(\beta)}, \quad (76)$$

we obtain the compact relation

$$\boxed{H_{\mathrm{MF}}(\beta) = H_{\mathrm{eff}}(\beta) + \frac{1}{\beta} \log Z_X(\beta) \, \mathbb{I}_S.} \quad (77)$$

Equivalently, if one chooses the conventional normalisation $\rho_S(\beta) \equiv \bar{\rho}_S(\beta)/Z_X(\beta)$, then $\rho_S(\beta) = e^{-\beta H_{\mathrm{MF}}(\beta)}$ exactly, and the scalar shift in Eq. (77) ensures $\mathrm{Tr}_S \rho_S(\beta) = 1$.

Closed-algebra example: qubit system (Pauli closure)

Assume the system is a qubit with Hamiltonian

$$H_Q = \frac{1}{2} \mathbf{h} \cdot \boldsymbol{\sigma}, \quad (78)$$

and the CL coupling operator is a Hermitian Pauli-direction

$$f = \mathbf{n} \cdot \boldsymbol{\sigma}, \quad (79)$$

with $\mathbf{h}, \mathbf{n} \in \mathbb{R}^3$. The Pauli product identity

$$(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = (\mathbf{a} \cdot \mathbf{b}) \, \mathbb{I} + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma} \quad (80)$$

implies the operator algebra closes on $\mathrm{span}\{\mathbb{I}, \boldsymbol{\sigma}\}$. Consequently, any reduced equilibrium operator can be written in Bloch form

$$\bar{\rho}_S(\beta) = r_0(\beta) \, \mathbb{I} + \mathbf{r}(\beta) \cdot \boldsymbol{\sigma}, \qquad r_0 > |\mathbf{r}|. \quad (81)$$

Step 1: adjoint-chain vectors. Define the nested-commutator chain

$$f^{(n)} \equiv \mathrm{ad}^n_{H_Q}(f) = [H_Q, [H_Q, \ldots [H_Q, f] \ldots]], \quad (82)$$

and write it in Pauli-vector form

$$f^{(n)} = \mathbf{u}_n \cdot \boldsymbol{\sigma}, \qquad \mathbf{u}_n \in \mathbb{R}^3. \quad (83)$$

For $H_Q = \frac{1}{2} \mathbf{h} \cdot \boldsymbol{\sigma}$ one has the recursion

$$\mathbf{u}_{n+1} = i \, \mathbf{h} \times \mathbf{u}_n, \qquad \mathbf{u}_0 = \mathbf{n}. \quad (84)$$

Equivalently, decomposing $\mathbf{n} = \mathbf{n}_{\parallel} + \mathbf{n}_{\perp}$ with respect to $\hat{\mathbf{h}} = \mathbf{h}/|\mathbf{h}|$, the chain closes on the three real directions $\{\mathbf{n}_{\parallel}, \mathbf{n}_{\perp}, \hat{\mathbf{h}} \times \mathbf{n}\}$.

Step 2: kernel moments and the quadratic influence operator Δ. Define the scalar kernel moments (bath data)

$$M_{nm}(\beta) \equiv \int_0^{\beta} d\tau \int_0^{\beta} d\tau' \; K(\tau - \tau') \, \frac{\tau^n}{n!} \frac{(\tau')^m}{m!}. \quad (85)$$

Then the (exact) Gaussian influence operator may be written in adjoint-chain form

$$\Delta(\beta) = \frac{1}{2} \sum_{n,m\geq 0} M_{nm}(\beta) \, f^{(n)} f^{(m)}. \quad (86)$$

Using Eq. (80) and $f^{(n)} = \mathbf{u}_n \cdot \boldsymbol{\sigma}$ gives a *closed* decomposition

$$\Delta(\beta) = \alpha(\beta) \, \mathbb{I} + \boldsymbol{\delta}(\beta) \cdot \boldsymbol{\sigma}, \quad (87)$$

with coefficients

$$\alpha(\beta) = \frac{1}{2} \sum_{n,m\geq 0} M_{nm}(\beta) \, \mathbf{u}_n \cdot \mathbf{u}_m, \quad (88)$$

$$\boldsymbol{\delta}(\beta) = \frac{i}{2} \sum_{n,m\geq 0} M_{nm}(\beta) \, \big(\mathbf{u}_n \times \mathbf{u}_m\big). \quad (89)$$

(Hermiticity of $\bar{\rho}_S$ enforces that the final α and $\boldsymbol{\delta}$ entering $\bar{\rho}_S$ are real; in practice this is ensured by the symmetry properties of M_{nm} and the structure of the $\mathbf{u}_n$.)

Step 3: reduced equilibrium operator in Bloch form. From the quenched construction one obtains

$$\bar{\rho}_S(\beta) = e^{-\beta H_Q}\, e^{\Delta(\beta)} = e^{\alpha(\beta)}\, e^{-\frac{\beta}{2}\mathbf{h}\cdot\boldsymbol{\sigma}}\, e^{\boldsymbol{\delta}(\beta)\cdot\boldsymbol{\sigma}}. \tag{90}$$

For any Pauli vector $\mathbf{x}$,

$$e^{\mathbf{x}\cdot\boldsymbol{\sigma}} = \cosh|\mathbf{x}|\,\mathbb{I} + \sinh|\mathbf{x}|\,\hat{\mathbf{x}}\cdot\boldsymbol{\sigma}. \tag{91}$$

Therefore the product in Eq. (90) can be multiplied explicitly to yield

$$\bar{\rho}_S(\beta) = r_0(\beta)\,\mathbb{I} + \mathbf{r}(\beta)\cdot\boldsymbol{\sigma}, \tag{92}$$

where, defining

$$\mathbf{x} \equiv -\frac{\beta}{2}\mathbf{h}, \qquad \mathbf{y} \equiv \boldsymbol{\delta}(\beta), \tag{93}$$

one finds

$$r_0(\beta) = e^{\alpha(\beta)}\Big(\cosh|\mathbf{x}|\cosh|\mathbf{y}| + (\hat{\mathbf{x}}\cdot\hat{\mathbf{y}})\,\sinh|\mathbf{x}|\,\sinh|\mathbf{y}|\Big), \tag{94}$$

$$\mathbf{r}(\beta) = e^{\alpha(\beta)}\Big(\hat{\mathbf{x}}\,\sinh|\mathbf{x}|\cosh|\mathbf{y}| + \hat{\mathbf{y}}\,\sinh|\mathbf{y}|\cosh|\mathbf{x}| + i(\hat{\mathbf{x}}\times\hat{\mathbf{y}})\,\sinh|\mathbf{x}|\,\sinh|\mathbf{y}|\Big). \tag{95}$$

(For the physical parameter regime yielding Hermitian $\bar{\rho}_S$, $\mathbf{r}(\beta)$ is real.)

Step 4: exact logarithm and the Hamiltonian of mean force. Any positive 2×2 operator in Bloch form $R = r_0\mathbb{I} + \mathbf{r}\cdot\boldsymbol{\sigma}$ has eigenvalues $\lambda_\pm = r_0 \pm |\mathbf{r}|$ and

$$\log R = \frac{1}{2}\log(\lambda_+\lambda_-)\,\mathbb{I} + \frac{1}{2}\log\left(\frac{\lambda_+}{\lambda_-}\right)\hat{\mathbf{r}}\cdot\boldsymbol{\sigma}. \tag{96}$$

Applying Eq. (96) to $\bar{\rho}_S(\beta)$ gives

$$H_{\mathrm{MF}}(\beta) = -\frac{1}{\beta}\log\left[\frac{\bar{\rho}_S(\beta)}{Z_X(\beta)}\right] = c(\beta)\,\mathbb{I} + \frac{1}{2}\,\mathbf{h}_{\mathrm{MF}}(\beta)\cdot\boldsymbol{\sigma}, \tag{97}$$

with

$$c(\beta) = -\frac{1}{2\beta}\log\big(\lambda_+(\beta)\lambda_-(\beta)\big) + \frac{1}{\beta}\log Z_X(\beta), \tag{98}$$

$$\mathbf{h}_{\mathrm{MF}}(\beta) = -\frac{1}{\beta}\log\left(\frac{\lambda_+(\beta)}{\lambda_-(\beta)}\right)\hat{\mathbf{r}}(\beta), \qquad \lambda_\pm(\beta) = r_0(\beta) \pm |\mathbf{r}(\beta)|. \tag{99}$$

Equations (88)–(99) constitute an exact closed-algebra construction of $H_{\mathrm{MF}}(\beta)$ for a qubit at arbitrary coupling strength: all bath dependence enters through the scalar moments $M_{nm}(\beta)$ (equivalently through K or $J(\omega)$), while all system dependence enters through the finite adjoint-chain vectors $\{\mathbf{u}_n\}$ determined by $(\mathbf{h},\mathbf{n})$.

Add-on: explicit H_{MF} for a qubit

We specialise to a qubit with

$$H_Q = \frac{1}{2}\,\mathbf{h}\cdot\boldsymbol{\sigma}, \qquad f = \mathbf{n}\cdot\boldsymbol{\sigma}, \qquad \mathbf{h},\mathbf{n}\in\mathbb{R}^3, \tag{100}$$

and define $\omega \equiv |\mathbf{h}|$ and $\hat{\mathbf{h}} \equiv \mathbf{h}/\omega$ (for $\omega \neq 0$). Decompose $\mathbf{n}$ into components parallel/perpendicular to $\hat{\mathbf{h}}$:

$$\mathbf{n}_\parallel \equiv (\mathbf{n}\cdot\hat{\mathbf{h}})\,\hat{\mathbf{h}}, \qquad \mathbf{n}_\perp \equiv \mathbf{n} - \mathbf{n}_\parallel, \qquad \mathbf{e}_2 \equiv \hat{\mathbf{h}}\times\mathbf{n}_\perp. \tag{101}$$

Interaction-picture coupling in closed form. Using the adjoint action of $\mathfrak{su}(2)$ (equivalently analytic continuation of the real-time rotation), the similarity transform $\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$ evaluates to

$$\tilde{f}(\tau) = \Big(\mathbf{n}_\parallel + \cosh(\omega\tau)\,\mathbf{n}_\perp + i\sinh(\omega\tau)\,\mathbf{e}_2\Big)\cdot\boldsymbol{\sigma}. \tag{102}$$

(For $\omega = 0$, one has $\tilde{f}(\tau) = f$.)

Mode projections. Recall the translation-invariant Matsubara representation

$$K(u) = \frac{1}{\beta}\sum_{\ell\in\mathbb{Z}}\kappa_\ell\, e^{i\nu_\ell u}, \qquad \nu_\ell = \frac{2\pi\ell}{\beta}, \qquad \kappa_{-\ell} = \kappa_\ell \in \mathbb{R}. \tag{103}$$

Define the mode-projected operators

$$\tilde{F}_\ell \equiv \int_0^\beta d\tau\, e^{i\nu_\ell\tau}\,\tilde{f}(\tau). \tag{104}$$

Using Eq. (102) and $e^{i\nu_\ell\beta} = 1$, we obtain the closed decomposition

$$\tilde{F}_\ell = \Big(A_\ell\,\mathbf{n}_\parallel + B_\ell\,\mathbf{n}_\perp + i\,C_\ell\,\mathbf{e}_2\Big)\cdot\boldsymbol{\sigma}, \tag{105}$$

with scalar coefficients

$$A_\ell \equiv \int_0^\beta d\tau\, e^{i\nu_\ell\tau} = \beta\,\delta_{\ell 0}, \tag{106}$$

$$B_\ell \equiv \int_0^\beta d\tau\, e^{i\nu_\ell\tau}\cosh(\omega\tau) = \frac{1}{2}\left(\frac{e^{\beta\omega}-1}{\omega+i\nu_\ell} + \frac{1-e^{-\beta\omega}}{\omega-i\nu_\ell}\right), \tag{107}$$

$$C_\ell \equiv \int_0^\beta d\tau\, e^{i\nu_\ell\tau}\sinh(\omega\tau) = \frac{1}{2}\left(\frac{e^{\beta\omega}-1}{\omega+i\nu_\ell} - \frac{1-e^{-\beta\omega}}{\omega-i\nu_\ell}\right). \tag{108}$$

For $\ell \neq 0$, $A_\ell = 0$ while B_ℓ and C_ℓ remain nonzero.

Exact influence operator $\Delta = \alpha\,\mathbb{I} + \boldsymbol{\delta}\cdot\boldsymbol{\sigma}$. From the sum-of-squares identity,

$$\Delta(\beta) = \frac{1}{2\beta}\sum_{\ell\in\mathbb{Z}}\kappa_\ell\,\tilde{F}_\ell\tilde{F}_\ell^\dagger, \tag{109}$$

and the Pauli product rule $(\mathbf{a}\cdot\boldsymbol{\sigma})(\mathbf{b}\cdot\boldsymbol{\sigma}) = (\mathbf{a}\cdot\mathbf{b})\mathbb{I} + i(\mathbf{a}\times\mathbf{b})\cdot\boldsymbol{\sigma}$, one finds

$$\Delta(\beta) = \alpha(\beta)\,\mathbb{I} + \boldsymbol{\delta}(\beta)\cdot\boldsymbol{\sigma}, \tag{110}$$

with *explicit* coefficients

$$\alpha(\beta) = \frac{1}{2\beta}\sum_{\ell\in\mathbb{Z}}\kappa_\ell\Big(|A_\ell|^2\,|\mathbf{n}_\parallel|^2 + |B_\ell|^2\,|\mathbf{n}_\perp|^2 + |C_\ell|^2\,|\mathbf{e}_2|^2\Big), \tag{111}$$

$$\boldsymbol{\delta}(\beta) = \frac{1}{2\beta}\sum_{\ell\in\mathbb{Z}}\kappa_\ell\Big(2\,\mathrm{Re}[B_\ell C_\ell^*]\,|\mathbf{n}_\perp|^2\,\hat{\mathbf{h}}\Big). \tag{112}$$

Here we used $\mathbf{n}_\parallel\times\mathbf{e}_2 = -(\mathbf{n}\cdot\hat{\mathbf{h}})\,\mathbf{n}_\perp$ and $\mathbf{n}_\perp\times\mathbf{e}_2 = |\mathbf{n}_\perp|^2\hat{\mathbf{h}}$, and the fact that the κ_ℓ are real and even in ℓ, so the non-Hermitian transverse components cancel in the ℓ-sum, leaving $\boldsymbol{\delta}(\beta)$ aligned with $\hat{\mathbf{h}}$. (For $\mathbf{n}_\perp = \mathbf{0}$ the coupling commutes with H_Q and $\boldsymbol{\delta} = 0$.)

It is convenient to define the two scalar bath-dressed qubit coefficients

$$\Xi_0(\beta) \equiv \frac{1}{2\beta}\sum_\ell \kappa_\ell|A_\ell|^2,$$
$$\Xi_c(\beta) \equiv \frac{1}{2\beta}\sum_\ell \kappa_\ell|B_\ell|^2,$$
$$\Xi_s(\beta) \equiv \frac{1}{2\beta}\sum_\ell \kappa_\ell|C_\ell|^2, \tag{113}$$

and

$$\Xi_{cs}(\beta) \equiv \frac{1}{\beta}\sum_\ell \kappa_\ell\,\mathrm{Re}[B_\ell C_\ell^*], \tag{114}$$

so that Eqs. (111)–(112) become

$$\alpha(\beta) = \Xi_0(\beta)\,|\mathbf{n}_\parallel|^2 + \Xi_c(\beta)\,|\mathbf{n}_\perp|^2 + \Xi_s(\beta)\,|\mathbf{e}_2|^2, \tag{115}$$

$$\boldsymbol{\delta}(\beta) = \frac{1}{2}\,\Xi_{cs}(\beta)\,|\mathbf{n}_\perp|^2\,\hat{\mathbf{h}}. \tag{116}$$

Since $|\mathbf{e}_2|^2 = |\mathbf{n}_\perp|^2$, the scalar part may be written as $\alpha(\beta) = \Xi_0|\mathbf{n}_\parallel|^2 + (\Xi_c + \Xi_s)|\mathbf{n}_\perp|^2$.

Collapse to a single exponent and take the logarithm. The reduced equilibrium operator is

$$\bar{\rho}_S(\beta) = e^{-\beta H_Q}e^{\Delta(\beta)} = e^{\alpha(\beta)}\,e^{\mathbf{x}\cdot\boldsymbol{\sigma}}\,e^{\mathbf{y}\cdot\boldsymbol{\sigma}},$$
$$\mathbf{x} \equiv -\frac{\beta}{2}\mathbf{h}, \qquad \mathbf{y} \equiv \boldsymbol{\delta}(\beta). \tag{117}$$

In $\mathfrak{su}(2)$ one has the exact composition rule

$$e^{\mathbf{x}\cdot\boldsymbol{\sigma}}\,e^{\mathbf{y}\cdot\boldsymbol{\sigma}} = e^{\mathbf{z}\cdot\boldsymbol{\sigma}}, \tag{118}$$

where $\mathbf{z}$ is determined by

$$\cosh|\mathbf{z}| = \cosh|\mathbf{x}|\cosh|\mathbf{y}| + (\hat{\mathbf{x}}\cdot\hat{\mathbf{y}})\,\sinh|\mathbf{x}|\sinh|\mathbf{y}|, \tag{119}$$

$$\hat{\mathbf{z}}\,\sinh|\mathbf{z}| = \hat{\mathbf{x}}\,\sinh|\mathbf{x}|\cosh|\mathbf{y}| + \hat{\mathbf{y}}\,\sinh|\mathbf{y}|\cosh|\mathbf{x}| + i(\hat{\mathbf{x}}\times\hat{\mathbf{y}})\,\sinh|\mathbf{x}|\sinh|\mathbf{y}|. \tag{120}$$

Thus

$$\bar{\rho}_S(\beta) = e^{\alpha(\beta)}\,e^{\mathbf{z}(\beta)\cdot\boldsymbol{\sigma}}, \tag{121}$$

and taking the logarithm is immediate:

$$\log\bar{\rho}_S(\beta) = \alpha(\beta)\,\mathbb{I} + \mathbf{z}(\beta)\cdot\boldsymbol{\sigma}. \tag{122}$$

Final explicit H_{MF}. Using $H_{\mathrm{MF}}(\beta) = -(1/\beta)\log[\bar{\rho}_S(\beta)/Z_X(\beta)]$, we obtain

$$\boxed{H_{\mathrm{MF}}(\beta) = \left(\frac{1}{\beta}\log Z_X(\beta) - \frac{\alpha(\beta)}{\beta}\right)\mathbb{I} - \frac{1}{\beta}\,\mathbf{z}(\beta)\cdot\boldsymbol{\sigma},} \tag{123}$$

where $\alpha(\beta)$ is given explicitly by Eqs. (111) or (115), $\boldsymbol{\delta}(\beta)$ by Eqs. (112) or (116), and $\mathbf{z}(\beta)$ is obtained from the closed composition relations (119)–(120) with $\mathbf{x} = -\frac{\beta}{2}\mathbf{h}$ and $\mathbf{y} = \boldsymbol{\delta}(\beta)$.

In the commuting limit $\mathbf{n}_\perp = \mathbf{0}$ one has $\boldsymbol{\delta}(\beta) = \mathbf{0}$ and $\mathbf{z} = \mathbf{x}$, so Eq. (123) reduces to $H_{\mathrm{MF}} = H_Q - (\kappa_0/2)f^2 + (1/\beta)\log Z_X\,\mathbb{I}$, as expected.

ALL-TO-ALL 2-LOCAL DESIGNABILITY FROM A FREE-SPIN CORE

The key consequence of the closure framework is constructive control: with a free-spin system Hamiltonian and a complete pair-resolved coupling basis, the Gaussian mean-force map can be used to synthesize finite-N all-to-all 2-local Hamiltonians. We make this statement precise and then benchmark it numerically via stochastic simulation of the free model.

Free-spin closure and operator content

We fix the bare system Hamiltonian to be strictly non-interacting,

$$H_S = \sum_{i=1}^{N}\left(\Omega_i^x X_i + \Omega_i^z Z_i\right). \tag{124}$$

For each site, ad_{H_S} acts only on the local Pauli triplet $\{X_i, Y_i, Z_i\}$, so the adjoint chain is finite-dimensional and closes sitewise. For pair-resolved channels

$$f_{ij}^{\alpha\beta} = u_{ij}^{\alpha\beta}\sigma_i^\alpha + v_{ij}^{\alpha\beta}\sigma_j^\beta, \qquad i<j,\ \alpha,\beta\in\{x,y,z\}, \tag{125}$$

products of adjoint-generated operators remain in the span of 0-local, 1-local, and 2-local Pauli strings. Thus, within this channel family, Gaussian influence terms do not generate operators beyond 2-local order.

Constructive all-to-all map

At fixed finite N, define the target coefficient set

$$H_{\mathrm{tar}} = \sum_i \mathbf{h}_i\cdot\boldsymbol{\sigma}_i + \sum_{i<j}\sum_{\alpha,\beta} J_{ij}^{\alpha\beta}\sigma_i^\alpha\sigma_j^\beta. \tag{126}$$

Using the pair-resolved basis above, the coefficient map is linear after flattening:

$$\mathbf{J} = A\,\boldsymbol{\theta}, \tag{127}$$

where $\mathbf{J}$ stacks all $J_{ij}^{\alpha\beta}$ and $\boldsymbol{\theta}$ stacks channel parameters. In the direct pair-resolved basis, A is full rank (identity up to ordering), so inversion is exact at finite N.

Important scope caveat: "arbitrary" means arbitrary within this finite-N, complete-basis setting. It is not a continuum-limit universality claim.

Numerical demonstration

We ran three complementary tests:

1. **Design-map inversion test** (all Pauli-pair channels active): random dense all-to-all targets at $N = 6, 8, 10, 12$.
2. **Physics agreement test**: stochastic free-model simulation versus exact thermal ED on $N = 6, 8, 10$ (two random instances each), using a dense all-to-all ZZ sector for stable reweighting.
3. **Scaling comparison**: stochastic runtime/memory versus quimb MPO+DMRG runtime/memory [40, 41] at $N = 12, 16, 20$.

Reported values below correspond to the production run ($\beta = 6$, $N \in \{6, 8, 10, 12, 16, 20\}$, samples $= 192$, τ-steps $= 96$, seeds $= 2$).

The reconstruction test shows full-rank maps with machine-precision inversion: all sampled instances satisfy $\mathrm{rank}(A) = \dim(\boldsymbol{\theta})$ and the relative reconstruction residual is numerically zero in the reported runs (Fig. 1).

For stochastic-vs-ED physics checks, the median absolute discrepancies across the six overlap runs are $|\Delta e| = 1.76 \times 10^{-2}$ (energy density), $|\Delta m_z| = 9.76 \times 10^{-2}$, and $|\Delta\overline{\langle Z_i Z_j\rangle}| = 3.61 \times 10^{-2}$ (Fig. 2). The energy-density relative error median is 8.75% at the present sampling/Trotter settings.

The scaling test shows the stochastic solver remains practical across the full all-to-all term count while keeping the same free-spin trajectory structure. In this run, stochastic wall time increased from 4.48 s ($N = 6$) to 53.4 s ($N = 20$), with peak memory from $\sim$ 195 MB to $\sim$ 741 MB. The MPO baseline (ground-state proxy) scaled from 0.45 s ($N = 12$) to 3.51 s ($N = 20$), with memory near 734–744 MB (Fig. 3).

NUMERICAL VALIDATION AGAINST WEAK/ULTRASTRONG PREDICTIONS

We benchmark the qubit predictions of Ref. [19] against exact finite-bath trace-out calculations over the full coupling range. The system Hamiltonian and coupling operator are

$$H_S = \frac{\omega_q}{2}\sigma_z, \qquad X = \cos\theta\,\sigma_z - \sin\theta\,\sigma_x, \tag{128}$$

with $\omega_q = 3$ and $\theta = 0.25$. The bath is discretised from $J(\omega) = Q\tau_c\omega e^{-\tau_c\omega}$ using $N_\omega = 3$ modes over $[0.5, 8.0]$, per-mode cutoff $n_{\max} = 4$, and $(Q, \tau_c) = (10, 1)$.

For each λ we compute $\rho_S^{\mathrm{exact}}(\lambda)$ from the global Gibbs state by exact diagonalisation and partial trace. We compare it against:

1. the bare Gibbs state $\tau_S \propto e^{-\beta H_S}$;
2. the ultrastrong projected state ρ_{US} from Eq. (7) of Ref. [19] (equivalently the qubit form, Eq. (8) there).

The primary diagnostics are

$$D_\tau = \frac{1}{2}\|\rho_S^{\mathrm{exact}} - \tau_S\|_1, \qquad D_{US} = \frac{1}{2}\|\rho_S^{\mathrm{exact}} - \rho_{US}\|_1, \tag{129}$$

plus the ℓ_1 coherence in the H_S eigenbasis.

The numerical results support the PRL predictions:

1. At $\lambda = 0$, D_τ is numerically zero ($\sim 10^{-16}$), as required.
2. As λ increases, ρ_S^{exact} departs from τ_S and approaches ρ_{US}. For $\beta = 1$, D_{US} drops from 0.1124 at $\lambda = 0$ to 1.99×10^{-3} at $\lambda = 8$, while D_τ increases to 0.1104.
3. Energetic coherence persists in the H_S basis at strong coupling, consistent with Eq. (8) in Ref. [19]: for $\beta = 1$, C_{H_S} increases from 0 at $\lambda = 0$ to 0.218 at $\lambda = 8$.

Figure 4 shows the full coupling scan at $\beta = 1$. Figure 5 shows the temperature dependence ($\beta = 0.5, 1, 2$): in all cases D_{US} is minimised at the largest simulated coupling and remains near 2×10^{-3} at $\lambda = 8$, while the crossover from Gibbs-like to ultrastrong-like behaviour shifts to smaller λ as β increases (approximately $0.6 \to 0.4 \to 0.2$).

A resolution check (low-resolution run: $N_\omega = 2$, $n_{\max} = 3$) changes the $\beta = 1$, $\lambda = 8$ diagnostics only slightly: $|D_{US}^{\mathrm{prod}} - D_{US}^{\mathrm{low}}| = 6.45 \times 10^{-4}$ and $|C_{H_S}^{\mathrm{prod}} - C_{H_S}^{\mathrm{low}}| = 1.25 \times 10^{-3}$, indicating that the qualitative conclusions are robust to the present discretisation.

DISCUSSION

This manuscript is positioned as the direct sequel to Ref. [1]. The earlier paper established the exact commuting Gaussian benchmark; the present one isolates the operator-algebraic obstruction that appears once $[H_Q, f] \neq 0$.

We have derived an exact operator reformulation of the reduced equilibrium object for a Gaussian bath, expressed both as a bilocal imaginary-time influence functional and as a quenched Gaussian-field average. The

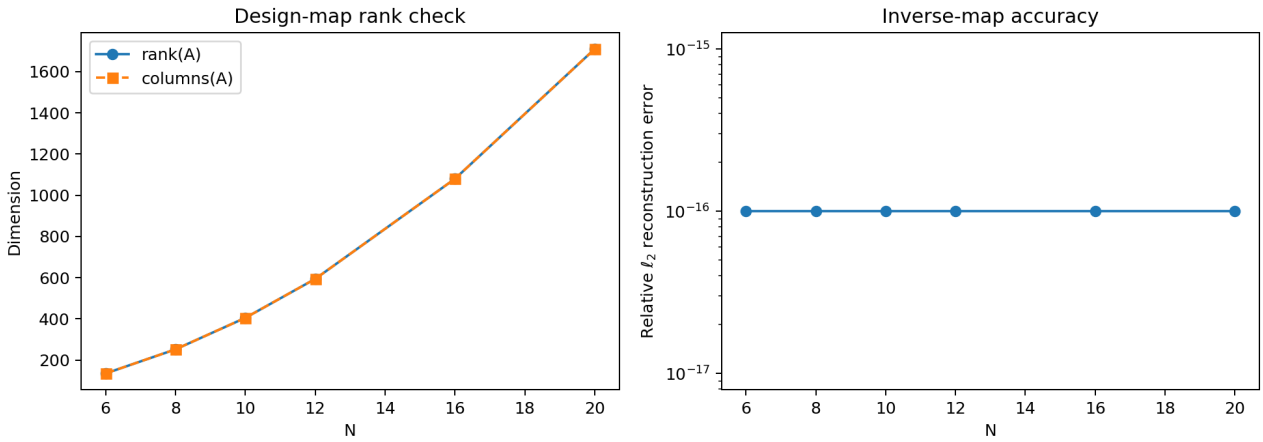

FIG. 1. All-to-all constructive-map diagnostics: rank(A) versus parameter-space dimension and inverse-map residual across sampled system sizes. The pair-resolved channel basis is full rank at finite N and reconstructs sampled target couplings to numerical precision.

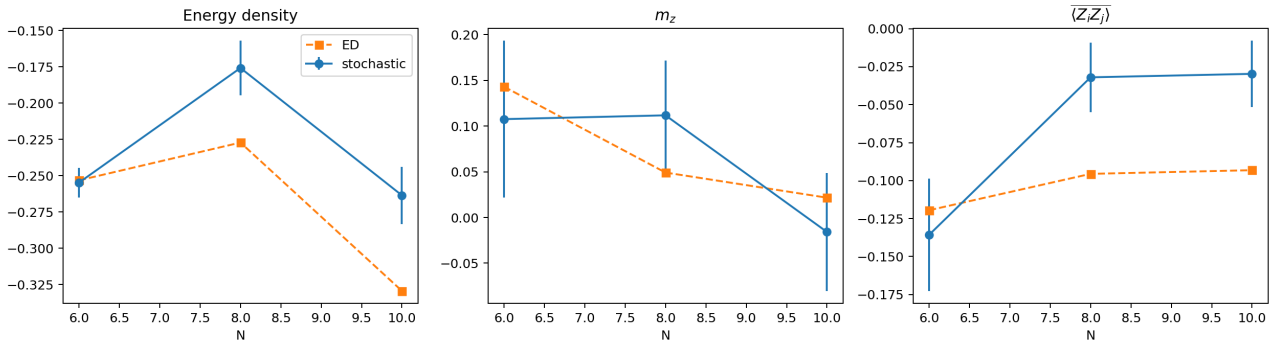

FIG. 2. Stochastic free-model benchmark against exact thermal ED on overlap sizes. Left: energy density. Middle: average m_z. Right: average all-to-all $\overline{\langle Z_i Z_j \rangle}$. Error bars show stochastic standard errors.

nonlocal structure in imaginary time is traced entirely to noncommutativity between H_Q and the coupling operator f, which forces the interaction-picture operator $\tilde{f}(\tau)$ to appear in time-ordered products. By expanding $\tilde{f}(\tau)$ in adjoint actions and isolating kernel moments, we obtained a purely algebraic representation that yields an exact closure criterion for locality of H_{MF}.

When the adjoint-generated operator algebra closes, the mean-force Hamiltonian is a finite operator polynomial and can be constructed via Magnus/BCH. When it does not, any local representation necessarily involves truncation or projection; no other approximation is introduced. Broader implications of these results are deferred to the next stage of the programme: non-Gaussian cumulant extensions and noncommuting finite-bath benchmarks.

The new designability section (Sec.) makes the main constructive consequence explicit for a free-spin core: with a complete pair-resolved channel basis, finite-N all-to-all 2-local coefficient maps are full rank and directly invertible. The numerical component there separates two roles: (i) reconstruction/rank diagnostics for the full all-to-all basis, and (ii) stochastic free-model benchmarking in a dense ZZ sector, where reweighting remains controlled enough to compare against exact thermal ED while also exposing scaling against an MPO/DMRG baseline.

The retained qubit benchmark (Sec.) continues to serve a different purpose: continuity with the weak/ultrastrong asymptotics of Ref. [19]. Together, the two numerical sections split validation into "known asymptotic physics recovery" (qubit) and "constructive many-body designability from free trajectories" (all-to-all construction).

Quenched Density and Imaginary-Time Evolution

This appendix makes explicit the operator identity underlying the quenched imaginary-time formulation used

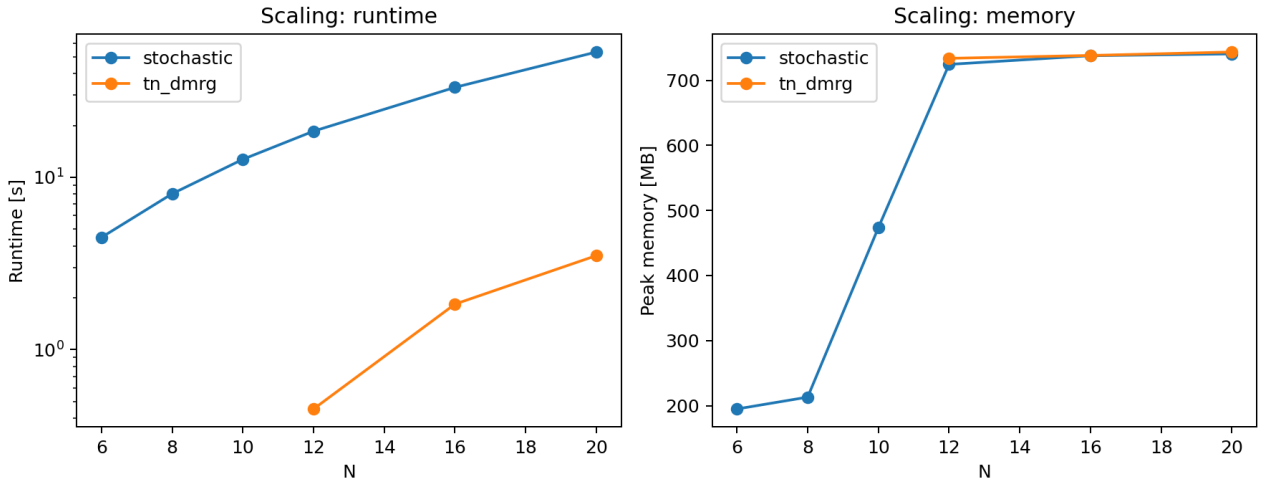

FIG. 3. Runtime and peak-memory scaling comparison between the stochastic free-model solver and a tensor-network MPO+DMRG baseline.

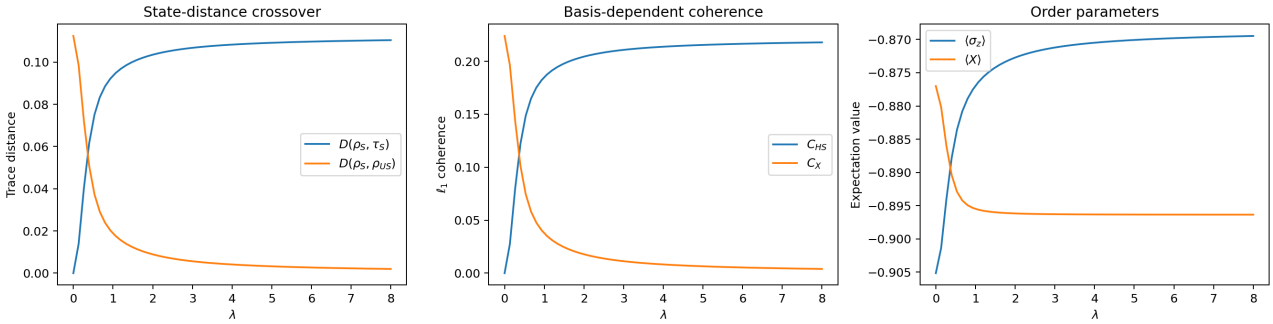

FIG. 4. Single-qubit benchmark at $\beta = 1$: trace-distance crossover, basis-dependent coherence, and expectation-value trends versus coupling strength λ. The exact finite-bath state moves away from the bare Gibbs state and towards the ultrastrong projected state as coupling grows.

in Sec. . Define a time-ordered exponential with a (generally) τ-dependent operator $H(\tau)$,

$$\rho(\tau) \equiv \mathcal{T}_\tau \exp\left(-\int_0^\tau d\tau'\, H(\tau')\right), \qquad \rho(0) = \mathbb{I}. \tag{130}$$

Standard differentiation identities for ordered exponentials imply

$$-\partial_\tau \rho(\tau) = H(\tau)\rho(\tau), \tag{131}$$

with the ordering built into $\rho(\tau)$; see, e.g., Refs. [42–44] for operator calculus and time-ordered exponentials.

In the present context one takes

$$H(\tau) = H_Q + \xi(\tau) f, \tag{132}$$

where the Gaussian field satisfies

$$\langle \xi(\tau)\xi(\tau')\rangle = K(\tau - \tau'). \tag{133}$$

The reduced equilibrium operator can be written as

$$\bar{\rho}_S = e^{-\beta H_Q} \langle \rho(\beta)\rangle_\xi, \tag{134}$$

which is the compact operator form of the quenched Gaussian representation used in the main text.

To connect Eq. (134) to an imaginary-time path integral, one discretizes $\tau \in [0, \beta]$, applies a Trotter (or Zassenhaus) factorization, and inserts resolutions of identity in the system coordinate basis. This yields the standard Euclidean path-integral expression for the canonical density operator, with the τ-dependent potential induced by the auxiliary field, as in the influence-functional derivation for quadratic baths and their stochastic unravellings [15, 16, 22, 23, 29].

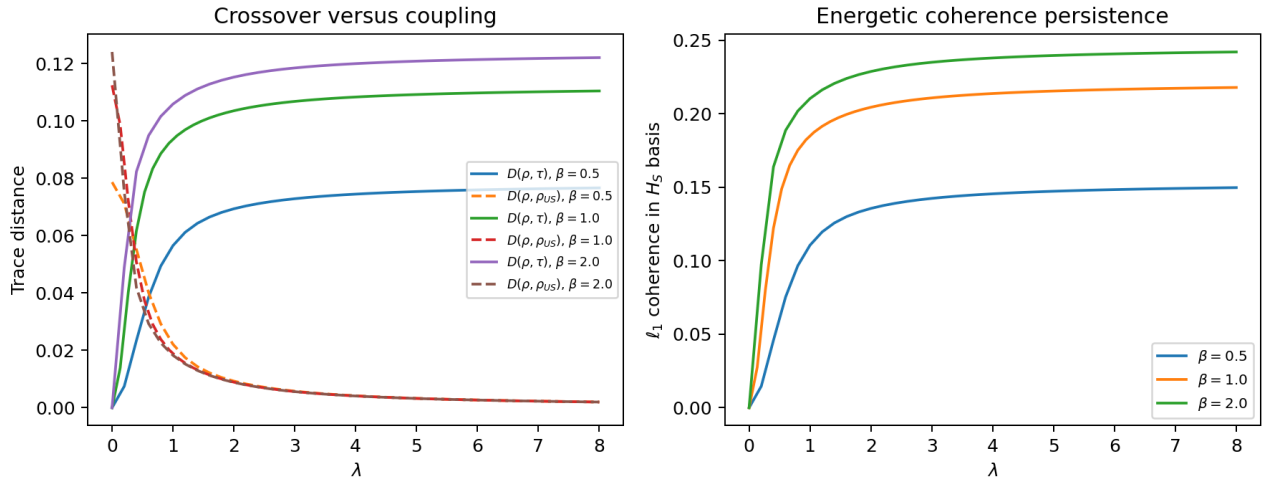

FIG. 5. Temperature comparison ($\beta = 0.5, 1, 2$): left panel shows D_τ (solid) and D_{US} (dashed), right panel shows energetic coherence in the H_S basis. Across temperatures, increasing coupling drives the exact state towards the ultrastrong predicted form while preserving nonzero energetic coherence.

Kernel Symmetry and Moment Relations

The bath kernel appearing in the bilocal influence functional is

$$K(\tau - \tau') = \mathrm{Tr}_B\left[\mathcal{T}_\tau \tilde{B}(\tau)\tilde{B}(\tau')\rho_B\right], \qquad \rho_B = \frac{e^{-\beta H_X}}{Z_B}. \tag{135}$$

For equilibrium baths, K depends only on the imaginary-time difference and is even under exchange of its arguments, implying

$$K(\tau - \tau') = K(\tau' - \tau), \qquad \mu_{nm} = \mu_{mn}, \tag{136}$$

$$\mu_{nm} = \frac{1}{n!m!}\int_0^\beta d\tau \int_0^\beta d\tau' \tau^n (\tau')^m K(\tau - \tau'). \tag{137}$$

In common quadratic-bath models the kernel is explicitly constructed from the bath spectral density and satisfies the Kubo-Martin-Schwinger periodicity in imaginary time, which can be used to re-express moment integrals in equivalent forms; see Refs. [16, 25, 26] for explicit constructions.

The moment expansion used in Sec. requires only these symmetry properties and the existence of the integrals defining μ_{nm}. No further approximation is introduced at this stage.

Derivation of the Influence Functional

In this appendix, we construct the influence functional formalism used in Sec. . Our goal is to derive the exact form of the reduced density operator $\bar{\rho}_S(\beta)$ by explicitly integrating out the harmonic bath, and to demonstrate that this leads directly to the stochastic unravelling employed in the main text.

Euclidean Path Integral Setup

We begin with the definition of the unnormalized reduced state,

$$\bar{\rho}_S(\beta) = \mathrm{Tr}_X\left[e^{-\beta H_{\mathrm{tot}}}\right]. \tag{138}$$

This trace can be represented as a Euclidean path integral. Let $|q\rangle$ and $|x\rangle = |x_1, x_2, \dots\rangle$ denote the position bases for the system and bath, respectively. The matrix element $\langle q|\bar{\rho}_S|q'\rangle$ involves a sum over all periodic paths $x(\tau)$ (where $x(0) = x(\beta)$) and open paths $q(\tau)$ (where $q(0) = q'$ and $q(\beta) = q$):

$$\langle q|\bar{\rho}_S|q'\rangle = \int_{q(0)=q'}^{q(\beta)=q} \mathcal{D}q(\tau) e^{-S_Q[q]/\hbar} \times \prod_k Z_k[q], \tag{139}$$

where S_Q is the Euclidean action of the isolated system, and $Z_k[q]$ is the partition function of the k-th oscillator in the presence of the external driving force $J_k(\tau) = -c_k f(q(\tau))$:

$$Z_k[q] = \oint \mathcal{D}x_k(\tau) \exp\left(-\frac{1}{\hbar}\int_0^\beta d\tau \times \left[\frac{m_k}{2}\dot{x}_k^2 + \frac{m_k\omega_k^2}{2}x_k^2 + c_k x_k f(q(\tau))\right]\right). \tag{140}$$

Note that the periodicity of the trace implies periodic boundary conditions for the bath paths $x_k(\tau)$.

Gaussian Integration

The functional integral for $Z_k[q]$ is Gaussian and can be evaluated exactly. It corresponds to the partition function of a forced harmonic oscillator. The result is expressible as the product of the free oscillator partition function, $Z_X^{(k)} = (2\sinh(\beta\hbar\omega_k/2))^{-1}$, and an exponential "influence phase" depending quadratically on the drive [29, 45]:

$$Z_k[q] = Z_X^{(k)} \exp\left(\frac{1}{2\hbar}\int_0^\beta d\tau \int_0^\beta d\tau' \times K_k(\tau-\tau') f(q(\tau)) f(q(\tau'))\right). \tag{141}$$

The kernel $K_k(\tau)$ is the equilibrium autocorrelation function of the coordinate x_k:

$$K_k(\tau-\tau') = c_k^2 \langle \mathcal{T}_\tau x_k(\tau) x_k(\tau')\rangle_0. \tag{142}$$

Summing over all modes k, the total influence functional is $\prod_k Z_k[q] = Z_X \exp(\Phi_{inf}[q])$, with

$$\Phi_{inf}[q] = \frac{1}{2\hbar}\int_0^\beta d\tau \int_0^\beta d\tau' K(\tau-\tau') \times f(q(\tau)) f(q(\tau')), \tag{143}$$

where $K(\tau) = \sum_k K_k(\tau)$ is the total force autocorrelation function.

From Non-local Action to Stochastic Average

Substituting this back into Eq. (139), the reduced density matrix becomes

$$\bar{\rho}_S = Z_X \int \mathcal{D}q\, e^{-S_Q[q]/\hbar} \exp\left(\Phi_{inf}[q]\right). \tag{144}$$

The term $\Phi_{inf}[q]$ is non-local in imaginary time, representing a self-interaction of the system mediated by the bath. To disentangle this, we use the Hubbard-Stratonovich transformation (the continuous analog of the Gaussian identity $e^{\frac{1}{2}A^2} \sim \int d\xi e^{-\frac{1}{2}\xi^2+\xi A}$). We introduce a real, auxiliary stochastic field $\xi(\tau)$ with zero mean and covariance

$$\langle \xi(\tau)\xi(\tau')\rangle_\xi = K(\tau-\tau'). \tag{145}$$

Using this field, we can rewrite the influence exponential as a stochastic average:

$$\exp\left(\Phi_{inf}[q]\right) = \left\langle \exp\left(\frac{1}{\hbar}\int_0^\beta d\tau \times \xi(\tau) f(q(\tau))\right)\right\rangle_\xi. \tag{146}$$

Inserting this identity into the path integral for $\bar{\rho}_S$, we can swap the order of the path integration over q and the stochastic average over ξ:

$$\bar{\rho}_S = Z_X \left\langle \int \mathcal{D}q \exp\left[-\frac{1}{\hbar}S_Q[q] + \frac{1}{\hbar}\int_0^\beta \xi f\right]\right\rangle_\xi. \tag{147}$$

The term in the angle brackets is exactly the path integral for a system evolving under the time-dependent Hamiltonian $H(\tau) = H_Q - \xi(\tau) f$. Thus, in operator language, we arrive at the exact stochastic representation:

$$\bar{\rho}_S(\beta) = Z_X \left\langle \mathcal{T}_\tau \exp\left(-\int_0^\beta d\tau \times [H_Q - \xi(\tau) f]\right)\right\rangle_\xi. \tag{148}$$

This confirms that the HMF can be constructed by averaging the non-unitary (imaginary-time) evolution of the system driven by colored Gaussian noise.

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